

1. Creating a New Protocol

Step 1 — New Protocol

After launching Brainstorm:

1. Navigate to *New Protocol*.
2. Set the following options:
 - *Default anatomy*: No
 - *Default Channel file*: No
3. Click *Create*.

2. Adding a Subject

Step 2 — New Subject

1. In the *Anatomy* tab, right-click on Protocol_name (anatomy).
2. Select *New Subject*.
3. Set the following options:
 - *Default anatomy*: No
 - *Default Channel file*: No
4. Click *Save*.

3. Setting the Subject Anatomy — Average Brain

Step 3 — Use Template Anatomy

1. In the *Anatomy* tab, right-click on Subject_name.
2. Select *Use Template*.
3. Choose *FsAverage* → *FsAverage_2020*.

3.2. (Optional) Change the Number of Cortex Vertices

Step 3.2 — Optional: Reducing the Number of Vertices

1. In the *Anatomy* tab, right-click on cortex_15002V.
2. Select *Less vertices*.
3. Enter the desired number of vertices, then click *Ok*.
4. Choose the *First method*, then click *Ok*.
5. The output file to retrieve at the end will be:
`tess_cortex_pial_low_(number_of_vertices)V.mat`

4. Importing EEG Electrodes

Step 4 — Default EEG Cap

1. Go to the *Functional Data (sorted by subjects)* tab.
2. Right-click on `Subject_name` and select *Use default EEG cap*.
3. Choose the electrode montage. Example configuration used:
 - *Colin27* → *Generic* → *ASA 10-05 343*

5. Generating BEM Surfaces

Step 5 — MRI Segmentation / BEM Surfaces

1. In the *Anatomy* tab, right-click on `Subject_name`.
2. Select *MRI segmentation* → *Generate BEM Surfaces*.
3. Set the options:
 - *Method*: `Brainstorm`
 - Choose the desired number of vertices for each compartment (scalp, outer skull, inner skull).
4. Click *Ok*.

6. Computing the Leadfield Matrix

Step 6 — Head Model (Leadfield)

1. Go to the *Functional Data (sorted by subjects)* tab.
2. Right-click on *Intra-subject* and select *Compute head model*.
3. Configure the parameters:
 - *Source space*: `Cortex Surface`
 - *Forward modelling method*: `EEG` and `OpenMEEG BEM`
4. Click *Ok*, then *Ok* again (leave default parameters).

7. Output Files

After the pipeline is complete, three files are of primary interest.

7.1. Leadfield Matrix and Channel File

Located in:

```
brainstorm_db\Protocol_name\data\Subject_name\@Intra-subjects\
```

File	Description
<code>headmodel.mat</code>	Contains the Leadfield (forward) matrix linking cortical sources to electrode measurements.
<code>channel_ASA_10-05_343.mat</code>	Contains the 3-D positions and labels of the 343 EEG electrodes.

7.2. Cortical Surface

Located in:

`brainstorm_db\Protocol_name\anat\Subject_name\`

File	Description
<code>tess_cortex_pial_low.mat</code>	Low-resolution cortical surface mesh (pial surface). This is the source space used for source localisation.